The Bootstrap

Estimating uncertainty by resampling the data that we already have

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Machine Learning Course

Outline

1. The Problem and the Idea
2. Worked Examples
3. Bootstrap Confidence Intervals
4. Properties of Bootstrap Samples
5. The Bootstrap in Machine Learning
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The Problem and the Idea

The problem: how reliable is my estimate?

- We observe one sample $\mathbf{x} = (x_1, \ldots, x_m)$, drawn i.i.d. from an unknown distribution F.
- We compute a statistic $\widehat{\theta} = s(\mathbf{x})$ estimating a parameter $\theta = t(F)$: a mean, a median, a correlation, a regression coefficient, a model's test accuracy, ...
- $\widehat{\theta}$ is a random variable: another sample would give another value. Its distribution is the **sampling distribution**.
- We want to know, for example:
  - **Standard error (SE)**: standard deviation of $\widehat{\theta}$ over repeated samples, $\mathrm{SE}(\widehat{\theta}) = \sqrt{\mathrm{Var}_F(\widehat{\theta})}$;
  - **Bias**: $\mathrm{Bias}(\widehat{\theta}) = \mathbb{E}_F[\widehat{\theta}] - \theta$;
  - **Confidence interval (CI)**: a data-based range $[L, U]$ with $P(L \le \theta \le U) \approx 1 - \alpha$ (e.g. 95%).

Difficulty

For the mean, $\mathrm{SE}(\bar{x}) = \sigma/\sqrt{m}$. For most other statistics (median, trimmed mean, ratio, a trained model, ...) there is **no simple formula**, and we cannot draw new samples from F.

The idea: the plug-in principle (Efron, 1979)

Replace the unknown F by the **empirical distribution** of the sample,

$$\widehat{F}_m \;:\; \text{probability } \tfrac{1}{m} \text{ on each } x_i,$$

$$\widehat{F}_m(t) = \frac{1}{m}\sum_{i=1}^{m} \mathbb{I}\{x_i \le t\}.$$

That is, the number of $x_i \le t$ divided by the total number of samples, m.

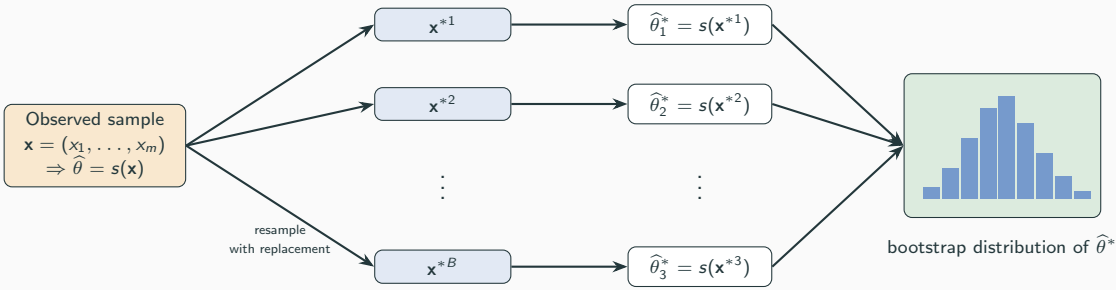
- Sampling from $\widehat{F}_m$ = drawing m points from $\mathbf{x}$ with replacement.
- Such a draw $\mathbf{x}^* = (x_1^*, \ldots, x_m^*)$ is a **bootstrap sample**.
- $\widehat{F}_m \to F$ as $m \to \infty$ (Glivenko–Cantelli), so the bootstrap world mimics the real one.

Two parallel worlds

Real world	Bootstrap world
F (unknown)	$\widehat{F}_m$ (known)
$\mathbf{x} \sim F$	$\mathbf{x}^* \sim \widehat{F}_m$
$\theta = t(F)$	$\widehat{\theta} = t(\widehat{F}_m)$
$\widehat{\theta} = s(\mathbf{x})$	$\widehat{\theta}^* = s(\mathbf{x}^*)$
$\widehat{\theta} - \theta$	$\widehat{\theta}^* - \widehat{\theta}$

The variability of $\widehat{\theta}^*$ around $\widehat{\theta}$ approximates the variability of $\widehat{\theta}$ around θ.

The bootstrap procedure



1: **for** $b = 1, \ldots, B$ **do** ▷ typically $B = 1000 - 10\,000$

2: draw indices $i_1, \ldots, i_m$ uniformly from $\{1, \ldots, m\}$ **with replacement**; $\mathbf{x}^{*b} = (x_{i_1}, \ldots, x_{i_m})$

3: compute $\widehat{\theta}^*_b = s(\mathbf{x}^{*b})$

4: **end for**

5: use the empirical distribution of $\widehat{\theta}^*_1, \ldots, \widehat{\theta}^*_B$ in place of the sampling distribution of $\widehat{\theta}$

Bootstrap estimates of standard error and bias

With $\bar{\theta}^* = \frac{1}{B}\sum_{b=1}^{B} \widehat{\theta}^*_b$:

$$\widehat{\mathrm{SE}}_{\mathrm{boot}} = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\widehat{\theta}^*_b - \bar{\theta}^*\right)^2} \qquad \widehat{\mathrm{Bias}}_{\mathrm{boot}} = \bar{\theta}^* - \widehat{\theta}$$

- Bias-corrected estimate: $\widehat{\theta}_{\mathrm{bc}} = \widehat{\theta} - \widehat{\mathrm{Bias}}_{\mathrm{boot}} = 2\widehat{\theta} - \bar{\theta}^*$ (use with care: it can increase variance).
- Two sources of error:
  - **statistical error**: $\widehat{F}_m \neq F$ — shrinks only as m grows;
  - **Monte Carlo error**: finite B — shrinks as B grows (cheap).
- As $B \to \infty$: the *ideal* bootstrap estimate. For the mean it equals the plug-in formula exactly:

$$\mathrm{SE}_{\widehat{F}_m}(\bar{x}^*) = \frac{\widehat{\sigma}}{\sqrt{m}}, \qquad \widehat{\sigma}^2 = \frac{1}{m}\sum_{i=1}^{m}(x_i - \bar{x})^2.$$

Worked Examples

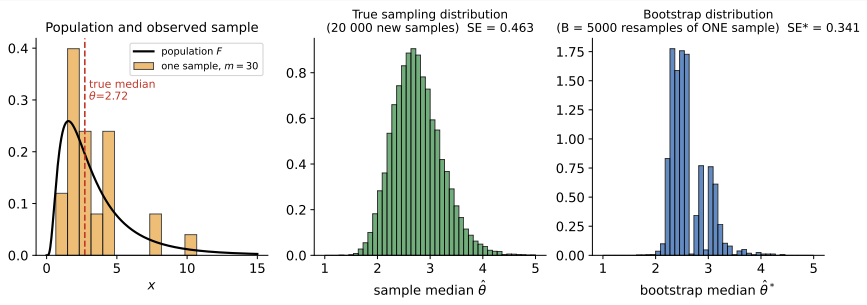
A toy example by hand ($m = 8$)

Observed sample: $\mathbf{x} = (2.1,\ 3.4,\ 1.8,\ 5.6,\ 2.9,\ 4.2,\ 3.1,\ 7.5)$, $\quad \bar{x} = 3.825$.

	indices drawn	bootstrap sample $\mathbf{x}^{*b}$	$\bar{x}^*_b$
$b = 1$	4, 5, 7, 8, 1, 2, 7, 8	5.6, 2.9, 3.1, 7.5, 2.1, 3.4, 3.1, 7.5	4.400
$b = 2$	2, 3, 7, 4, 3, 7, 3, 4	3.4, 1.8, 3.1, 5.6, 1.8, 3.1, 1.8, 5.6	3.275
$b = 3$	6, 5, 1, 1, 7, 7, 7, 5	4.2, 2.9, 2.1, 2.1, 3.1, 3.1, 3.1, 2.9	2.938
$\vdots$			$\vdots$

- Some points appear several times (7.5 twice in $b = 1$; 1.8 three times in $b = 2$), others not at all.
- With $B = 10\,000$: $\widehat{\mathrm{SE}}_{\mathrm{boot}}(\bar{x}) = 0.637$.
- Check: plug-in formula $\widehat{\sigma}/\sqrt{m} = 0.630$; usual formula $s/\sqrt{m} = 0.674$ (with $m - 1$).
  The bootstrap reproduces the known answer — and works just as easily for any other statistic.

Example: standard error of the median



Population: log-normal, $\log X \sim \mathcal{N}(1, 0.75^2)$, true median $\theta = e \approx 2.72$. One sample of $m = 30$ gives $\widehat{\theta} = 2.51$.

- The middle panel needs the unknown population; the right panel uses only the one sample.
- $\widehat{\mathrm{SE}}_{\mathrm{boot}} = 0.34$ vs. true $\mathrm{SE} = 0.46$: the right order of magnitude from one sample of 30. The histogram is lumpy because a resample's median can only take a few of the original values.

Bootstrap Confidence Intervals

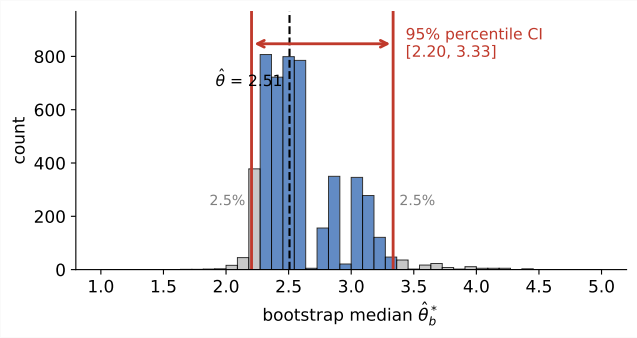
Three common bootstrap confidence intervals

Let $\widehat{\theta}^*_{(\alpha)}$ denote the α-quantile of $\widehat{\theta}^*_1, \ldots, \widehat{\theta}^*_B$. A $(1-\alpha)$ CI for θ:

Method	**Interval**	**Median example ($\widehat{\theta} = 2.51$)**
Normal	$\widehat{\theta} \pm z_{1-\alpha/2}\, \widehat{\mathrm{SE}}_{\mathrm{boot}}$	[1.84, 3.17]
Percentile	$\left[\widehat{\theta}^*_{(\alpha/2)},\ \widehat{\theta}^*_{(1-\alpha/2)}\right]$	[2.20, 3.33]
Basic (pivotal)	$\left[2\widehat{\theta} - \widehat{\theta}^*_{(1-\alpha/2)},\ 2\widehat{\theta} - \widehat{\theta}^*_{(\alpha/2)}\right]$	[1.68, 2.81]

- **Normal**: assumes $\widehat{\theta}$ is approximately Gaussian.
- **Percentile**: simplest; respects skewness and range restrictions.
- **Basic**: uses $\widehat{\theta}^* - \widehat{\theta} \approx \widehat{\theta} - \theta$, so it reflects the quantiles around $\widehat{\theta}$.
- Refinements (BCa, studentized bootstrap) correct bias and skewness; they are more accurate but costlier.

The percentile interval

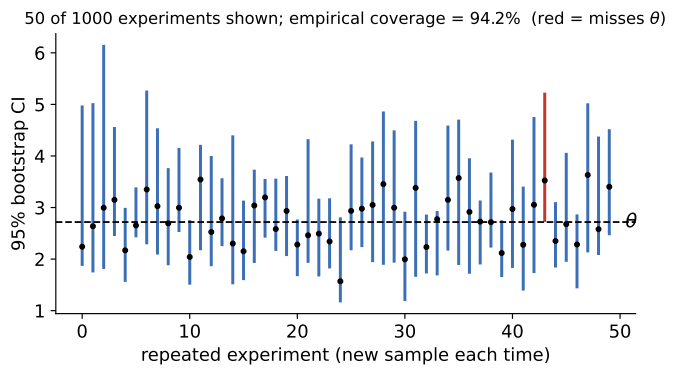


- Sort the $B = 5000$ bootstrap medians.
- Cut off 2.5% on each side (grey bars).
- The remaining range is the 95% CI:

$$[\widehat{\theta}^*_{(0.025)},\ \widehat{\theta}^*_{(0.975)}] = [2.20,\ 3.33].$$

- With $B = 5000$ these are the 125th and 4875th sorted values.

Does it work? Checking coverage



Repeat the whole experiment 1000 times: draw a new sample of $m = 30$, build a 95% percentile interval from $B = 1000$ resamples, and check whether it contains the true median $\theta = e$. Empirical coverage: 94.2%, close to the nominal 95%. Any single interval may miss (red), but the *procedure* is calibrated.

Properties of Bootstrap Samples

How many distinct points does a bootstrap sample contain?

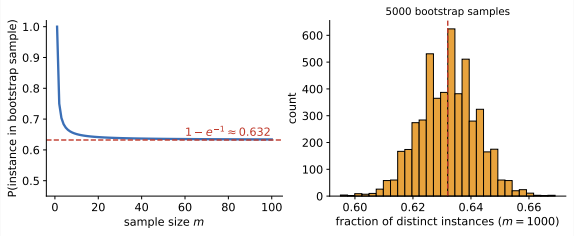
Probability that instance i is *never* drawn in m draws:

$$\left(1 - \frac{1}{m}\right)^m \xrightarrow[m\to\infty]{} e^{-1} \approx 0.368.$$

So each instance appears with probability

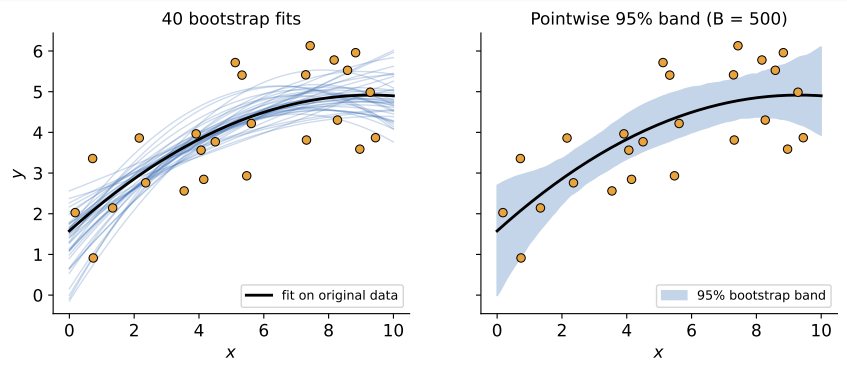
$$1 - \left(1 - \frac{1}{m}\right)^m \approx 0.632.$$

- A bootstrap sample contains about 63.2% distinct instances; the rest are duplicates.
- The $\approx 36.8\%$ left out are the **out-of-bag** (OOB) instances.



The Bootstrap in Machine Learning

Example: uncertainty of a fitted regression curve



Resample the (x_i, y_i) *pairs* with replacement, refit a quadratic model $\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x + \widehat{\beta}_2 x^2$ each time, and take pointwise 2.5% and 97.5% quantiles of the B fitted curves. The band is wide where data are sparse (near the edges).

Where the bootstrap appears in machine learning

Bagging and Random Forests

Train one model per bootstrap sample and aggregate:

$$\widehat{f}_{\text{bag}}(\mathbf{x}) = \frac{1}{B}\sum_{b=1}^{B} \widehat{f}^{*b}(\mathbf{x}).$$

Averaging reduces the variance of unstable learners such as deep trees.

Out-of-bag error

Each instance is OOB for $\approx 37\%$ of the models; predicting it with those models gives a free validation estimate.

Model evaluation

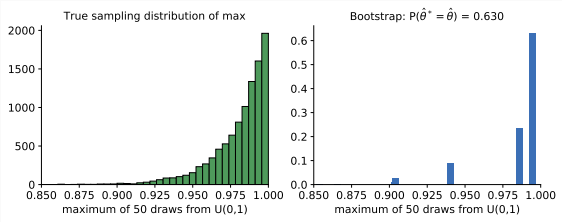
- CI for a test metric: resample the *test set* and recompute accuracy / AUC.
- Comparing two models: bootstrap the *difference* of their metrics on the same resampled test sets.
- The **.632 estimator** mixes training error and OOB error:

$$\widehat{\mathrm{Err}}_{.632} = 0.368\,\overline{\mathrm{err}}_{\text{train}} + 0.632\,\widehat{\mathrm{Err}}_{\text{OOB}}.$$

Limitations

When the bootstrap fails

- **Dependent data**: plain resampling destroys the order of a time series or the structure of grouped data.
  $\Rightarrow$ use a *block* or *cluster* bootstrap.
- **Very small** m: $\widehat{F}_m$ is a poor stand-in for F.
- **Extreme statistics** (max, min, extreme quantiles): a resample can never exceed the sample maximum.
- Heavy tails with infinite variance.



$\widehat{\theta} = \max_i x_i$, $x_i \sim \mathcal{U}(0,1)$, $m = 50$. The bootstrap puts $1 - (1 - \frac{1}{m})^m \approx 63\%$ of its mass on the single value $\widehat{\theta}$ itself: it cannot mimic the true, continuous distribution.

Summary

- The bootstrap estimates the **sampling distribution** of any statistic $\widehat{\theta}$ from a single sample by resampling it with replacement.
- Plug-in principle: treat the empirical distribution $\widehat{F}_m$ as if it were the population F.
- From $\widehat{\theta}_1^*, \ldots, \widehat{\theta}_B^*$ we obtain

$$\widehat{\mathrm{SE}}_{\mathrm{boot}}, \qquad \widehat{\mathrm{Bias}}_{\mathrm{boot}} = \bar{\theta}^* - \widehat{\theta}, \qquad \text{normal / percentile / basic CIs.}$$

- Each bootstrap sample contains $\approx 63.2\%$ of the distinct instances; the rest are out-of-bag.
- In ML: bagging, Random Forests, OOB error, confidence intervals for test metrics.
- It needs i.i.d. data and a reasonably large m, and fails for extreme statistics such as the maximum.

Questions?